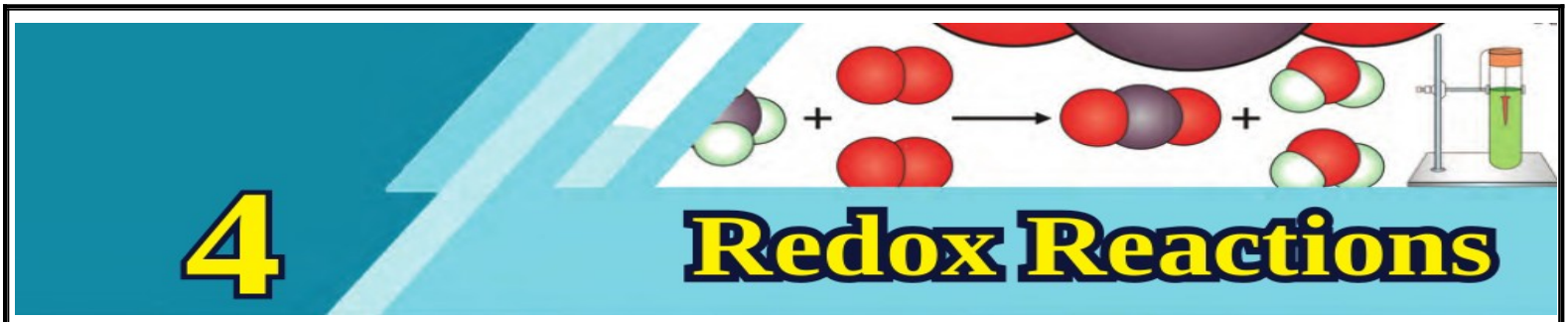


# 4 Redox Reactions

Law of conservation of mass

In a chemical reaction, the total mass of the reactants will be equal to the total mass of the products.

Balancing a chemical equation is the method of equalising the number of the same type of atoms in both the reactants and the products. The equation thus obtained is known as a balanced chemical equation.

1. Magnesium + Oxygen → Magnesium oxide

Step 1 $Mg + O_2 \rightarrow MgO$

Step 2 $Mg + O_2 \rightarrow \underline{2}MgO$

Step 3 $\underline{2}Mg + O_2 \rightarrow 2MgO$

Balanced chemical equation $\mathbf{2Mg + O_2 \rightarrow 2MgO}$

- Balance the chemical equations given below and record them in science diary.

$H_2 + I_2 \rightarrow HI$

$Na + H_2O \rightarrow NaOH + H_2$

$Mg + HCl \rightarrow MgCl_2 + H_2$

$H_2 + I_2 \rightleftharpoons 2HI$ , $2Na + 2H_2O \longrightarrow 2NaOH + H_2$

$Mg + 2HCl \longrightarrow MgCl_2 + H_2$

The process involving loss of electrons in a chemical reaction is called oxidation . $Na \longrightarrow Na^+ + 1e^-$

The process involving gaining of electrons in a chemical reaction is called reduction. $Cl + 1e^- \longrightarrow Cl^-$

Generally, non metals undergo reduction during chemical reactions.

Positive ions also can gain electrons and change into atoms. Such reactions are also reduction reactions. $Ag^+ + 1e^- \rightarrow Ag$

negative ions lose electrons and change into atoms. Such reactions are also oxidation reactions. $I^- \rightarrow I + 1e^-$

The species that helps oxidation in a chemical reaction is the Oxidising agent . The oxidising agent gets reduced in a chemical reaction.

$$2Na + Cl_2 \longrightarrow 2NaCl$$

Which atom undergoes oxidation ?

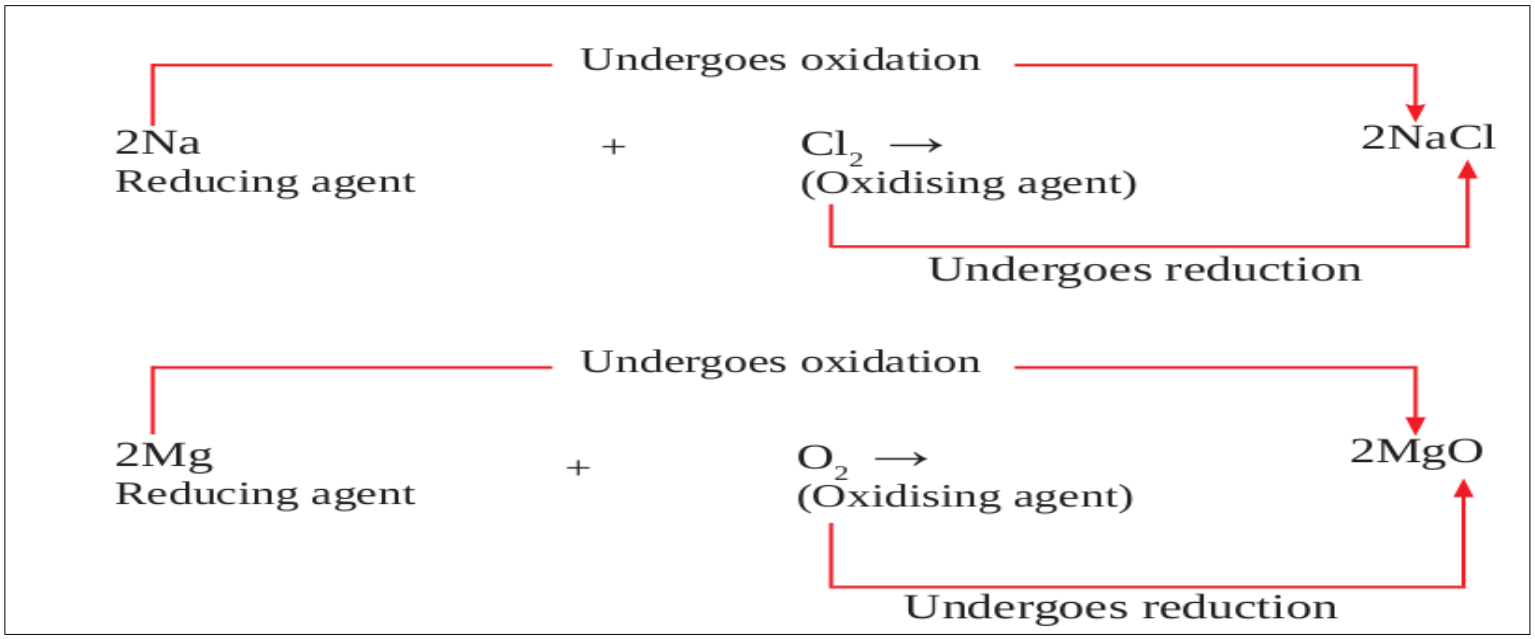
Sodium loses electron and undergoes oxidation.

Which atom supports oxidation? (sodium/chlorine)

Chlorine gains electron and helps oxidation.

The species that helps reduction is the reducing agent. The

reducing agent gets oxidised in a chemical reaction.




- Analyse the chemical reactions given below and complete the table.

1) $Mg + F_2 \rightarrow MgF_2$

2) $Ca + Cl_2 \rightarrow CaCl_2$

3) $4Fe + 3O_2 \rightarrow 2Fe_2O_3$

| Sl.No. | Equation of oxidation | Equation of reduction | Oxidising agent | Reducing agent |
|---|---|---|---|---|
| 1 | $Mg \rightarrow Mg^{2+} + 2e^-$ | F +1e ---> F- | F | ...Mg.... |
| 2 | Ca --->Ca+2.+2e- | $Cl + 1e^- \rightarrow Cl^-$ | .....Cl... | Ca |
| 3 | $Fe \rightarrow Fe^{3+} + 3e^-$ | O +1e ---> O- | O | ....Fe.... |

| Element | Atomic number | Electron configuration | Valency |
|---|---|---|---|
| Sodium (Na) | 11 | 2, 8 1 | 1 |
| Potassium (K) | 19 | 2, 8, 8, 1 | 1 |
| Fluorine (F) | 9 | 2, 7 | 1 |
| Chlorine (Cl) | 17 | 2, 8, 7 | 1 |
| Magnesium (Mg) | 12 | 2, 8, 2 | 2 |
| Calcium (Ca) | 20 | 2, 8, 8, 2 | 2 |
| Oxygen (O) | 8 | 2, 6 | 2 |
| Aluminium (Al) | 13 | 2, 8, 3 | 3 |

If all the chemical bonds in a compound are considered to be ionic, the charge formed on each atom is considered as its oxidation number or oxidation state.

The oxidation number of magnesium is +2 and that of oxygen is −2 in magnesium oxide (MgO). What do you understand from this?

Magnesium looses 2 electrons and oxygen gains 2 electrons.

The sum of the oxidation numbers of all atoms in compound is zero. In element molecules, electrons are equally shared by the atoms. So, at elemental state the oxidation number considered to be zero.

oxidation and reduction take place simultaneously. Such reactions are called redox reactions.

In a redox reaction, oxidising agent undergoes reduction and reducing agent undergoes oxidation.

| Element | Oxidation number |
|---|---|
| H | +1 |
| Na | +1 |
| K | +1 |
| Ca | +2 |
| Al | +3 |
| F | –1 |
| Cl | –1 |
| Br | –1 |
| I | –1 |
| O | –2 |

Find out the oxidation number of chromium in $Cr_2O_3$.

Find out the oxidation number of manganese (Mn) in the following compounds and record it in your science diary.

(Hint: Oxidation number of O = –2, K = +1.)

a) $MnO_2$ b) $Mn_2O_7$ c) $KMnO_4$

Oxidation number of Cr=+3

a)+4 b)+7 c)+7

What is the oxidation number of sodium and chlorine in their elemental state? Zero.

Oxidation number increases during oxidation reactions.

Reduction reactions involve a decrease in oxidation number.

The unbalanced chemical equation regarding the formation of ammonia from nitrogen and hydrogen is given below.

$$N_2 + H_2 \rightarrow NH_3$$

a) Balance the chemical equation.

b) Find out the total number of atoms of the same type in both the reactants and the products.

c) If 28 g of nitrogen combines with 6 g of hydrogen, find out the mass of ammonia formed. (Hint : Atomic mass H=1u N=14u)

a)$N_2$ +3$H_2$ --->2$NH_3$ b)nitrogen=2, hydrogen=6 3)34

$C + 4HNO_3 \rightarrow 2H_2O + CO_2 + 4NO_2$

a) Find out and mark the oxidation number of carbon in this reaction.

b) What happens to the oxidation number of carbon in this reaction?

c) What happens to carbon-oxidation or reduction?

d) What are the oxidising and reducing agents in this reaction?

a)$C^0 + 4HNO_3 \rightarrow 2H_2O + C^{+4}O_2 + 4NO_2$

b)increases c)oxidation d)oxidising agent—nitric acid

reducing agent—carbon

Find out the oxidation number of sulphur in the following compounds.

(Hint : Oxidation number H = +1, O = –2)

a) $SO_2$ b) $SO_3$ c) $H_2SO_3$ d) $H_2SO_4$

a)+4 b)+6 c)+4 d)+6

Certain statements are given below. Write whether they are true or false.

The process involving an increase in oxidation number is oxidation.

The process involving a decrease in oxidation number is oxidation.

In a chemical reaction, oxidising agent undergoes reduction.

In a chemical reaction, oxidising agent undergoes oxidation.

a)true b)false c)true d)false

Balance the chemical equations given below.

a) $SO_2 + O_2 \rightarrow SO_3$ b) $H_2O_2 \rightarrow H_2O + O_2$

c) $CH_4 + O_2 \rightarrow H_2O + CO_2$ d) $Fe + HCl \rightarrow FeCl_2 + H_2$

a)$2SO_2 + O_2 \rightarrow 2SO_3$ b)$2H_2O_2 \rightarrow 2H_2O + O_2$

c)$CH_4 + 2O_2 \rightarrow 2H_2O + CO_2$ d) $Fe + 2HCl \rightarrow FeCl_2 + H_2$

Two chemical reactions are given below. Find out the oxidation number of atoms and check whether these reactions are redox reactions.

a) $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ b) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

a)$Ca^{+2}\ C^{+4}\ O_3^{-2} + 2H^{+1}\ Cl^{-1} \rightarrow Ca^{+2}\ Cl_2^{-1} + H_2^{+1}O^{-2} + C^{+4}\ O_2^{-2}$

this is not a redox reaction.

$Zn^0 + 2H^{+1}\ Cl^{-1} \rightarrow Zn^{+2}\ Cl_2^{-1} + H_2^{\ 0}$ -this is redox reaction.

The gaseous fuel carbon monoxide burns in oxygen to form carbon dioxide.

a) Write the balanced equation of this chemical reaction.

b) Is this a redox reaction? Why?

c) What is the oxidising agent in this reaction? What is the reducing agent?

a)$2CO + O_2 \rightarrow 2CO_2$ b)yes,bcz both oxidation and reduction take place.

c)oxidising agent—oxygen reducing agent---carbon monoxide.

Analyse the chemical equations given below and find out whether they are redox reactions.

a) $NaOH + HCl \rightarrow NaCl + H_2O$

b) $H_2S + Cl_2 \rightarrow 2HCl + S$

a)no b)yes

Analyse the chemical equation given below.

$$Ca + 2HCl \rightarrow CaCl_2 + H_2$$

a) Mark the oxidation number of atoms before and after the chemical reaction.
b) Which atom undergoes oxidation?
c) Which atom undergoes reduction?
d) What are the oxidising and reducing agents?

a) $Ca^{0} + 2H^{+1} Cl^{-1} \rightarrow Ca^{+2} Cl_2^{-1} + H_2^{0}$ b)calcium c)hydrogen

d)oxdising agent---hydrogenchloride reducing agent---calcium

). A chemical reaction is given in the concept map below. Find out the oxidation number of each atom. On the basis of this, fill up the blanks.

(Hint : Valency S = 2, Fe = 2)

